

BCNmodular

MAESTRO

Probabilistic Voice Router, Arranger, Mixer, Sequencer & Trigger Generator

User Manual v0 | Module version 2.1.0

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1. Introduction

Maestro is an arrangement tool for VCV Rack 2 that brings controlled randomness to your modular patches. Instead of building fully deterministic sequences, Maestro acts as an intelligent conductor, deciding which voices play at each moment based on weighted probability, density control, and musical timing.

In modular synthesis, achieving musical structure while preserving randomness typically requires combining many modules. Maestro consolidates this into a single module: connect your voices, set their relative probabilities, and let Maestro decide who plays at each evaluation point.

1.1 The core concept

Maestro works by periodically evaluating which of your 6 channels should be active. This evaluation happens every N bars (you decide), and the result is a probabilistic selection based on:

- How many voices you want active (Density)
- How random the selection should be (Randomness)
- The individual probability weight of each channel (Prob knobs)

This creates evolving arrangements that have musical structure but never repeat exactly — the key balance between composition and improvisation that makes modular synthesis so compelling.

1.2 What Maestro is NOT

- It is not a sequencer in the traditional sense — it does not play notes
- It is not a mixer — it does not control volume levels
- It is not a random module — its randomness is musically controlled

1.3 System requirements

- VCV Rack 2.x (Free or Pro)
- Compatible with all operating systems supported by VCV Rack

2. Installation

Maestro is available through the official VCV Library.

2.1 From the VCV Library (recommended)

1. Open VCV Rack 2
2. Click Library in the top menu
3. Search for BCNmodular or Maestro
4. Click Subscribe and then Update plugins
5. Restart VCV Rack

2.2 From GitHub (manual)

Download the latest .vcvplugin file for your platform from:

github.com/santifort-commits/BCNmodular/releases

Place the file in your VCV Rack plugins folder:

- Windows: Documents\Rack2\plugins
- Mac: ~/Documents/Rack2/plugins
- Linux: ~/Documents/Rack2/plugins

3. Quick Start

This section gets you up and running with a basic Maestro patch in 5 minutes.

3.1 Your first Maestro patch

Step 1: Add Maestro to your rack

- Right-click on an empty rack area and search for Maestro
- Place it anywhere in your rack

Step 2: Connect a clock

- Connect your master clock output to the CLOCK input (top left of Maestro)
- Maestro needs a clock signal to count bars and trigger evaluations

Step 3: Connect your voices

- Connect the gate or audio output of your first voice to CH1 INPUT
- Connect CH1 OUTPUT back to your voice's destination (voice gate, mixer input, VCA input, etc.)
- Repeat for up to 6 voices

Step 4: Set basic parameters

- ACTIVE TRACKS: set to the number of voices you have connected (e.g. 3)
- TRACK DENS: set to how many voices you want active at once (e.g. 2)
- LENGTH: set to 1 or 2 bars for quick evaluation

Step 5: Start your clock and listen

- Maestro will now periodically decide which voices are active
- Watch the LEDs to see which channels are open (green) or closed (red)

TIP Start with Gate mode (G) for all channels. Once you understand the basic behaviour, experiment with Fade mode (F) for audio channels and Trigger mode (T) for triggering events. Use Gate mainly to control gate signals and Fade for audio signals.

4. Panel Overview

Maestro has 18HP width and is organized in three zones:

- Row 1 (Global controls): Clock, density, and output settings
- Row 2 (Timing controls): Length, skip, and fade parameters
- Channel rows (x6): Individual channel controls

4.1 Global controls (Row 1)

Control	Type	Description
CLOCK	Input	Clock/trigger input. Drives the bar counter. Connect to your master clock.
ACTIVE TRACKS	Knob	Number of channels participating in evaluation (1 to 6). Channels above this number are always inactive.
TRACK DENS	Knob	Base number of active voices (0 to ACTIVE TRACKS). This is the target number of channels Maestro will try to keep active.
DENS CV ATTVERT	Knob	Attenuverter for the Density CV input. Center = no effect. Right = full positive effect. Left = full negative (inverted) effect.
DENS CV	Input	CV input for density modulation (0-10V). Scaled by the attenuverter.
DETERM/RANDOM	Knob	Randomness amount. Fully left (D) = deterministic, always exactly TRACK DENS voices. Fully right (R) = maximum gaussian distribution.
OUT CH. AMOUNT	Output	CV output proportional to the number of currently active voices (0-10V). Configurable in context menu.

4.2 Timing controls (Row 2)

Control	Type	Description
RESET	Input	Resets the bar counter to zero (default) or forces an immediate evaluation. Configurable in context menu.
LENGTH	Knob	Evaluation period in bars: 1, 2, 4, 8 or 16 bars. After this many bars, Maestro re-evaluates which voices are active.
LEN CV	Input	CV input for length. When connected, overrides the LENGTH knob.
SKIP PROB	Knob	Probability of skipping an evaluation (0 = never skip, 1 = always skip). When an evaluation is skipped, the current state is preserved.
SKIP CV	Input	CV input for skip probability. Behavior depends on Skip CV mode (context menu).
FADE IN	Knob	Fade-in time for channels in Fade mode (0 to 10 seconds).

FADE OUT	Knob	Fade-out time for channels in Fade mode (0 to 10 seconds). Longer times preserve natural reverb tails.
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4.3 Per-channel controls (x6)

Control	Type	Description
LABEL	Text	Editable 4-character label. Double-click to edit. Saved with the patch.
INPUT	Input	Signal input: gate, clock, audio or trigger. If unconnected, defaults to 1V (configurable in context menu).
PROB	Knob	Base probability weight for this channel (0 to 1). Higher values make this channel more likely to be selected.
PROB CV ATTVERT	Knob	Attenuverter for the Prob CV input (bidirectional, -1 to +1).
PROB CV	Input	CV modulation for the channel probability. Scaled by the attenuverter.
T/G/F	Switch	Channel mode: T = Trigger, G = Gate, F = Fade. See section 5 for details.
LED	Light	RGB status indicator. Colors depend on the channel mode. See section 5.
OUT	Output	Signal output. The input signal is passed (or blocked) according to the channel state and mode.

5. Channel Modes (T/G/F)

Each channel has an independent mode switch with three positions. This is one of Maestro's most powerful features, as different signal types benefit from different modes.

5.1 Gate mode (G)

Gate mode provides instant on/off switching. When a channel is selected by Maestro, the input signal passes through immediately. When deselected, the signal is blocked immediately.

Best for:

- Clock and gate signals going to sequencers
- Controlling external switches or VCAs
- Any signal where instant response is desired

LED behavior in Gate mode:

- Green: channel is active, signal passing through
- Red: channel is inactive, signal blocked

5.2 Fade mode (F)

Fade mode applies a smooth amplitude envelope when opening or closing a channel. The fade-in and fade-out times are set globally by the FADE IN and FADE OUT knobs.

Best for:

- Audio signals (avoids clicks and pops)
- Preserving natural reverb tails when closing a channel
- Smooth transitions between voices

LED behavior in Fade mode:

- Green: channel is fully open
- Yellow: fade transition in progress (opening or closing)
- Red: channel is fully closed

TIP Use long FADE OUT times (3-10 seconds) when your voices have natural reverb or delay tails. This prevents abrupt cuts that would sound unnatural.

5.3 Trigger mode (T)

Trigger mode sends a short 10V pulse each time the channel is selected by an evaluation. This is different from Gate mode because the signal is only 'on' for a very brief moment (1-10ms, configurable), not for the entire evaluation period.

Best for:

- Firing burst generators and envelope generators
- Triggering samplers and drum modules
- Creating probabilistic percussion patterns
- Sending one-shot events to any trigger-sensitive module

LED behavior in Trigger mode:

- Blue (constant): channel is in trigger mode, waiting to be selected
- White flash: trigger fired (300ms flash)

Important notes about Trigger mode:

- When no input is connected, Trigger mode always outputs 10V (regardless of the Default input voltage setting)
- If an input is connected, the trigger pulse uses the input voltage
- All selected channels fire their trigger simultaneously
- The trigger duration can be adjusted in the context menu (1ms, 2ms, 5ms default, 10ms)

TIP If a module does not respond to Maestro triggers, try increasing the trigger duration to 10ms in the context menu. Some modules need longer pulses. Some modules are sensitive to different voltages so this could be adjusted also in the contextual menu if needed.

6. Context Menu Options

Right-click on Maestro to access advanced settings. All settings are saved with your patch.

Option	Values	Description
Beats per bar	2, 3, 4, 5, 6, 7, 8 (default: 4)	Sets the time signature. LENGTH is counted in bars, and each bar has this many beats (clock pulses).
Min active voices	0 to 6 (default: 0)	Minimum number of channels that must be active at all times. Set to 1 or more to prevent complete silence.
Default input voltage	1V (default) or 10V	Voltage sent to output when no input is connected. Use 1V for gates/triggers, 10V for CV control.
Active output CV mode	Proportional / Absolute	Proportional: OUT CH. AMOUNT scales to ACTIVE TRACKS (e.g. 3 of 3 = 10V). Absolute: always scales to 6 (e.g. 3 of 6 = 5V).
Trigger duration	1ms, 2ms, 5ms (default), 10ms	Duration of the trigger pulse in Trigger mode. Increase if modules do not respond reliably.
Skip CV mode	Probabilistic / Binary	Probabilistic: SKIP CV sets the probability (0-10V = 0-100%). Binary: below 1V = never skip, above 1V = always skip.
Skip mode	Global / Per channel	Global: entire evaluation is skipped or not. Per channel: each channel independently decides whether to keep its current state.
Reset input mode	Reset bars / Force evaluate	Reset bars: RESET input resets the bar counter to zero. Force evaluate: RESET input triggers an immediate evaluation without resetting the counter.

7. Typical Use Cases

7.1 Arrangement tool

Connect the output of multiple sequencers or voice modules to Maestro inputs. Set LENGTH to 2 or 4 bars. Maestro will periodically decide which voices are active, creating evolving arrangements that never repeat exactly.

Recommended settings:

- All gate channels in Gate mode (G) and audio channels in Fade mode (F)
- DETERM/RANDOM fully left for predictable voice counts
- Individual PROB knobs to weight your favourite voices
- LENGTH: 2-4 bars

7.2 Performance tool

Use Maestro as a live performance controller. Connect a MIDI CC to the DENS CV input and use the attenuverter to control how much the external signal affects density. During performance, you can manually turn the TRACK DENS knob to bring the amount of voices up and down.

Recommended settings:

- Reset input mode: Force evaluate (context menu)
- Connect a momentary button or footswitch to RESET
- Press RESET to trigger a new arrangement at any moment
- Skip CV mode: Binary — use a gate signal to freeze the current arrangement

7.3 Probabilistic trigger sequencer

Set all channels to Trigger mode and leave inputs unconnected. Connect outputs to burst generators, envelope generators, or drum modules. Each evaluation, Maestro decides which 'instruments' fire their trigger.

Recommended settings:

- All channels in Trigger mode (T)
- Skip mode: Per channel (context menu) — each channel independently may or may not fire
- Individual PROB knobs for each drum voice
- LENGTH: 1 bar for drum patterns, 2-4 bars for longer phrases
- Trigger duration: 10ms if any module does not respond

7.4 Smooth audio mixer

Connect audio outputs from different voices to Maestro inputs. Use Fade mode with long fade times for smooth crossfades. Maestro becomes an intelligent audio switcher that transitions gracefully between sounds.

Recommended settings:

- All channels in Fade mode (F)
- FADE IN: 0.1 to 0.5 seconds
- FADE OUT: 2 to 10 seconds (preserve reverb tails)
- TRACK DENS: 1 or 2 (only a few voices at once for clarity)

7.5 Self-regulating patch

Connect the OUT CH. AMOUNT output back to the DENS CV input through an attenuverter. This creates a feedback loop where the current density influences future density decisions, creating interesting evolving behaviors.

- Active output CV mode: Proportional (context menu)
- Set DENS CV ATTVERT to a small negative value
- This creates a 'regression to the mean' effect — if too many voices are active, fewer will be selected next time

8. Tips & Tricks

TIP Use channel labels (double-click) to name your voices: KICK, BASS, PAD, LEAD, etc. This makes complex patches much easier to understand at a glance.

TIP Set DETERM/RANDOM fully left for exact voice counts. Turn it right gradually to introduce variation in the number of active voices.

TIP In 3/4 or 6/8 time, set Beats per bar to 3 or 6 in the context menu to ensure Maestro evaluates on musically correct boundaries.

TIP Use MIN ACTIVE VOICES in the context menu when you always need at least one voice playing (e.g. bass or kick drum should never stop).

TIP Longer FADE OUT times (5-10 seconds) work beautifully with reverb-heavy pads — the reverb tail fades naturally as the next voice comes in.

TIP In Force evaluate mode, the RESET input becomes a performance controller. Map it to a MIDI button or footswitch to manually trigger arrangement changes.

TIP Mix channel modes: use Gate for clock signals, Fade for audio, and Trigger for percussion. Different modes for different signal types in the same patch.

TIP The ACTIVE CV output can modulate other parameters — for example, connect it to a filter cutoff to automatically open the filter when more voices are active.

9. Technical Reference

9.1 Signal levels

Signal	Range	Notes
CLOCK input	> 1V = high	Rising edge triggers beat count
RESET input	> 1V = trigger	Rising edge only
DENS CV input	0 to 10V	Scaled by attenuverter
LEN CV input	0 to 10V	Overrides LENGTH knob when connected
SKIP CV input	0 to 10V (probabilistic) / 0 or 10V (binary)	See Skip CV mode
PROB CV inputs	0 to 10V	Scaled by per-channel attenuverter
OUT CH. AMOUNT	0 to 10V	Proportional or absolute (see context menu)
Trigger output (T mode)	10V pulse	Duration: 1-10ms (configurable)
Gate output (G mode)	Input voltage	Passes or blocks input signal
Fade output (F mode)	0 to input voltage	Smooth amplitude envelope

9.2 Polyphony

All 6 channels support polyphonic signals. The number of polyphonic channels is automatically detected from the input and preserved at the output. Fade levels are applied equally to all polyphonic channels.

9.3 The evaluation algorithm

When an evaluation is triggered, Maestro performs the following steps:

1. Calculate target voices: $\text{TRACK DENS} + (\text{DENS CV} \times \text{attenuverter})$, clamped to [Min active voices, ACTIVE TRACKS].
2. Apply randomness: if $\text{DETERM}/\text{RANDOM} > 0$, the target is varied using a gaussian distribution with sigma proportional to the randomness setting.
3. Weighted random selection: channels are selected using weighted random sampling without replacement. Each channel's weight is its PROB knob value plus $(\text{PROB CV} \times \text{attenuverter})$.
4. Apply skip probability: in Global mode, the entire evaluation may be skipped. In Per channel mode, each channel independently may retain its previous state.

9.4 Saving and loading

The following settings are saved with your VCV Rack patch (.vcv file):

- All knob positions
- All channel mode switches (T/G/F)
- All channel labels
- Context menu settings: beats per bar, min active voices, default input voltage, active output CV mode, trigger duration, skip CV mode, skip mode, reset input mode

10. Changelog

Version 2.1.0

- Added Trigger mode (T) per channel: sends a 10V pulse when selected
- RGB LEDs: blue = trigger mode, white flash on trigger, green = open, yellow = fading, red = closed
- Configurable trigger duration: 1ms, 2ms, 5ms (default), 10ms via context menu
- Skip mode: Global or Per channel (context menu)
- Skip CV mode: Probabilistic or Binary (context menu)
- Reset input mode: Reset bars or Force evaluate (context menu)
- Fixed: simultaneous triggers now fire correctly on all selected channels

Version 2.0.0

- Initial public release

11. License & Contact

Maestro is released under the GPL-3.0-or-later license. The source code is available at:
github.com/santifort-commits/BCNmodular

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If you find Maestro useful and want to support its development:
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Bug reports and feature requests: please open an issue on GitHub.